

PROBLEM LOOKUP — READ FIRST

|                                                 |                                           |
|-------------------------------------------------|-------------------------------------------|
| Question asks for...                            | Go to                                     |
| power density $S(R, \theta)$ from a dipole      | DIPOLE TYPE $\rightarrow$ HERTZIAN or ARB |
| HPBW $\beta$ from $F(\theta)$                   | BEAM PARAM step 2                         |
| solid angle $\Omega_p$ / directivity $D$        | BEAM PARAM step 3–4                       |
| $\theta_{\max}$ , $S_{\max}$ , plot $F(\theta)$ | ARB-LENGTH DIPOLE                         |
| effective area $A_e$ / ratio $A_e/A_p$          | EFFECTIVE AREA                            |
| $P_r$ given (link)                              | FRIIS FORMULA                             |

Step 0 always: compute  $\lambda = c/f$ . Step 1: classify dipole by  $l/\lambda$ .

DIPOLE TYPE BY LENGTH

| Compute first $\lambda = c/f$ , then ratio $l/\lambda$ |                  |                                                               |
|--------------------------------------------------------|------------------|---------------------------------------------------------------|
| $l/\lambda$                                            | Type             | Use formula                                                   |
| $< 1/50$                                               | Hertzian (short) | HERTZIAN $S(R, \theta)$ , $D = 1.5$                           |
| $= 1/4$                                                | Quarter-wave     | ARB formula, $\pi l/\lambda = \pi/4$                          |
| $= 1/2$                                                | Half-wave        | half-wave shortcut, $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ |
| $= 1$                                                  | Full-wave        | ARB, $D = 2.41$ , $R_{\text{rad}} = 199 \Omega$               |
| other                                                  | Arbitrary        | ARB formula                                                   |

If  $l/\lambda < 1/50$  STOP and use Hertzian. Do NOT plug small  $l$  into the arbitrary formula — you’ll get the wrong shape.

HERTZIAN DIPOLE —  $S(R, \theta)$

|                                                                                                                                                                                                                                                                                                         |                                          |
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| Far-field E,H phasors                                                                                                                                                                                                                                                                                   | Lec 15 eq 19                             |
| $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi} \frac{e^{-jkR}}{R} \sin \theta$ , $\vec{H}_\phi = \vec{E}_\theta / \eta_0$                                                                                                                                                                               |                                          |
| Power density                                                                                                                                                                                                                                                                                           | $l/\lambda < 1/50$ ; far field           |
| $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\max} \sin^2 \theta$ (W/m <sup>2</sup> )                                                                                                                                                                                    |                                          |
| $k = 2\pi/\lambda$ (rad/m); $I_0$ peak current (A); $l$ length (m); $R$ obs. dist (m).                                                                                                                                                                                                                  |                                          |
| Max direction                                                                                                                                                                                                                                                                                           | broadside $\theta = \pi/2$               |
| Total radiated power                                                                                                                                                                                                                                                                                    | integrate over $4\pi$                    |
| $P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda}\right)^2$ (W)                                                                                                                                                                                                                        |                                          |
| Directivity                                                                                                                                                                                                                                                                                             | exact $D_{\text{Hertz}} = 1.5$ (1.76 dB) |
| HW4 P1: $l = 1\text{m}$ , $f = 1\text{MHz}$ , $I_0 = 12\text{A}$ , $R = 5\text{km}$ , $\theta = 45^\circ$ . $\lambda = 300\text{m}$ , $l/\lambda = 1/300 < 1/50$ ✓. $S = \frac{(120\pi)^2 (2\pi/300)^2 (144)(1)}{32\pi^2 (5 \times 10^3)^2} \cdot \frac{1}{2} = 1.51 \times 10^{-9}$ W/m <sup>2</sup> . |                                          |

ARBITRARY-LENGTH DIPOLE —  $S(\theta)$

|                                                                                                                                                       |                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Power density                                                                                                                                         | any $l$ , at distance $R$                                            |
| $S(\theta) = \frac{15I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$      |                                                                      |
| Normalized pattern                                                                                                                                    | dimensionless $F(\theta) = S(\theta)/S_{\max}$                       |
| Half-wave                                                                                                                                             | $l = \lambda/2$ $\pi l/\lambda = \pi/2$ , shortcut form              |
| $S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$                                                         |                                                                      |
| $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ , $\theta_{\max} = \pi/2$ .                                                                                 |                                                                      |
| Quarter-wave                                                                                                                                          | $l = \lambda/4$ $\pi l/\lambda = \pi/4$ , $\cos(\pi/4) = \sqrt{2}/2$ |
| $\frac{S(\theta)}{S_0} = \left[ \frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$ , $S_0 = \frac{15I_0^2}{\pi R^2}$ |                                                                      |
| $S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$                                                                  |                                                                      |
| HW4 P3: $\theta_{\max} = \pi/2$ (broadside, same as half-wave); $S_{\max} = 0.0858 S_0$ .                                                             |                                                                      |

BEAM PARAMETERS — HPBW,  $\Omega_p$ ,  $D$

|                                                                                                                                            |                                                                                        |
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| Step 1 — Normalize                                                                                                                         | always start here                                                                      |
| $F(\theta) = S(\theta)/S_{\max}$                                                                                                           |                                                                                        |
| Step 2 — HPBW                                                                                                                              | $\beta$ solve $F(\theta_{1/2}) = 0.5$ , then $\beta = 2\theta_{1/2}$ (symmetric beams) |
| Gaussian shortcut: $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$ .                                                  |                                                                                        |
| HW4 P2: $F = e^{-20\theta^2}$ , $a = 20 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/20} = 0.186$ rad, $\beta = 0.372$ rad = $21.31^\circ$ .     |                                                                                        |
| Step 3 — Solid angle $\Omega_p$ (sr)                                                                                                       |                                                                                        |
| $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$                                                                  |                                                                                        |
| If $F$ has no $\phi$ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$ .                                                    |                                                                                        |
| Most integrals have NO closed form. Use Wolfram / numerical integration.                                                                   |                                                                                        |
| HW4 P2: $\Omega_p = 2\pi \int_0^\pi e^{-20\theta^2} \sin \theta d\theta = 0.156$ sr (numeric).                                             |                                                                                        |
| Step 4 — Directivity                                                                                                                       |                                                                                        |
| $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$                                                                    |                                                                                        |
| Alt. if two orthogonal HPBWs known: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$ . HW4 P2: $D = 4\pi/0.156 = 80.55$ (19.06 dB), very directive. |                                                                                        |
| Step 5 — Gain & efficiency $\xi = \text{radiation eff.}$ , $0 < \xi \leq 1$                                                                |                                                                                        |
| $G = \xi D = \frac{4\pi R^2 S_{\max}}{P_t}$ $G_{\text{dB}} = 10 \log_{10} G$                                                               |                                                                                        |
| Lossless / ideal antenna $\Rightarrow \xi = 1$ , $G = D$ .                                                                                 |                                                                                        |

$R_{\text{rad}}$ , EFFICIENCY, POWER SPLIT

|                                                                                                                                      |                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Radiation resistance                                                                                                                 | $\Omega$ , equivalent ckt. load that absorbs $P_{\text{rad}}$ |
| $R_{\text{rad}} = \frac{2P_{\text{rad}}}{I_0^2}$ $R_{\text{loss}} = \frac{2P_{\text{loss}}}{I_0^2}$                                  |                                                               |
| Input power split                                                                                                                    | $P_t$ from generator                                          |
| $P_t = P_{\text{rad}} + P_{\text{loss}}$ $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$                                          |                                                               |
| Radiation efficiency                                                                                                                 | two equivalent forms                                          |
| $\xi = \frac{P_{\text{rad}}}{P_t} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}$                                         |                                                               |
| Hertzian: $R_{\text{rad}} = 80\pi^2 (l/\lambda)^2$ small. Half-wave: $R_{\text{rad}} = 73 \Omega$ matches 50/75 $\Omega$ lines well. |                                                               |

COMMON DIPOLE PARAMETERS

| Type          | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}}$ ( $\Omega$ ) |
|---------------|------|-----------------|-------------------------------|
| Isotropic     | 1    | 0               | —                             |
| Hertzian      | 1.5  | 1.76            | $80\pi^2 (l/\lambda)^2$       |
| Half-wave     | 1.64 | 2.15            | 73                            |
| $l = \lambda$ | 2.41 | 3.82            | 199                           |

Half-wave is the default “dipole antenna” unless stated otherwise.

EFFECTIVE AREA  $A_e$

|                                                                                                                                                                                                                                 |                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Definition                                                                                                                                                                                                                      | antenna’s RX cross section (m <sup>2</sup> ) |
| $A_e = \frac{\lambda^2 D}{4\pi}$ (m <sup>2</sup> )                                                                                                                                                                              |                                              |
| Physical cross section                                                                                                                                                                                                          | wire dipole, diameter $d$                    |
| $A_p = l d = \frac{\lambda}{2} d$ (half-wave)                                                                                                                                                                                   |                                              |
| Inverse: $D$ from $A_e$                                                                                                                                                                                                         | $D = \frac{4\pi A_e}{\lambda^2}$             |
| HW4 P4: half-wave, $f = 100\text{MHz}$ , $d = 2\text{cm}$ . $\lambda = 3\text{m}$ , $D = 1.64 \Rightarrow A_e = \frac{9(1.64)}{4\pi} = 1.17$ m <sup>2</sup> . $A_p = (1.5)(0.02) = 0.03$ m <sup>2</sup> . Ratio 39.2 $\times$ . |                                              |
| Thin wires: $A_e \gg A_p$ . Antenna captures field over $\sim \lambda$ region.                                                                                                                                                  |                                              |

POWER FLOW IDENTITIES

|                                                                        |  |
|------------------------------------------------------------------------|--|
| $P_{\text{rad}} = S_{\max} \Omega_p R^2 = \frac{4\pi R^2 S_{\max}}{D}$ |  |
| $S_{\max} = \frac{P_{\text{rad}} D}{4\pi R^2}$ (W/m <sup>2</sup> )     |  |
| Isotropic radiator: $S = P_{\text{rad}}/(4\pi R^2)$ (no $D$ factor).   |  |

FRIIS TRANSMISSION FORMULA

|                                                                                                                                                                                                                                                                      |                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Far-field link budget                                                                                                                                                                                                                                                | $P_t$ tx, $P_r$ rx, range $R$ |
| $P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$ (W)                                                                                                                                                                                                        |                               |
| $G_t, G_r$ linear gains; $\lambda = c/f$ ; $R$ in m; $P_t, P_r$ in W.                                                                                                                                                                                                |                               |
| Solve for $R$ max range                                                                                                                                                                                                                                              |                               |
| $R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$ (m)                                                                                                                                                                                                        |                               |
| Solve for $P_r$ using $A_e$ equivalent form                                                                                                                                                                                                                          |                               |
| $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$ , $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$                                                                                                                                                                                    |                               |
| Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$ , $P_r = G_t G_r P_t / L_{fs}$ .                                                                                                                                                                                 |                               |
| Use linear $G$ . Convert dB first: $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$ .                                                                                                                                                                                        |                               |
| HW4 P5: half-wave TX, $P_t = 1\text{kW}$ , $f = 50\text{MHz}$ , RX $G_r = 3\text{dB} = 2$ , $R = 30\text{km}$ . $\lambda = 6\text{m}$ , $G_t = 1.64$ . $P_r = (1.64)(2)\left(\frac{6}{4\pi \times 10^4}\right)^2 10^3 = 8.3 \times 10^{-7}$ W (0.83 $\mu\text{W}$ ). |                               |

dB  $\leftrightarrow$  LINEAR

| $G_{\text{dB}} = 10 \log_{10} G_{\text{lin}}$ |        | $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$ |        |
|-----------------------------------------------|--------|------------------------------------------|--------|
| dB                                            | linear | dB                                       | linear |
| 0                                             | 1      | 10                                       | 10     |
| 3                                             | 2      | 13                                       | 20     |
| 6                                             | 4      | 20                                       | 100    |
| 7                                             | 5      | 30                                       | 1000   |
| −3                                            | 0.5    | −10                                      | 0.1    |

For E-field or amplitude use  $20 \log$  not  $10 \log$  (power is squared).

Memorize: 3 dB  $\Leftrightarrow \times 2$ , 10 dB  $\Leftrightarrow \times 10$ . Add dB = multiply linear.

ANTENNA INTEGRATION TIPS

$\Omega_p$  integrals usually have no closed form.  
Wolfram: `2*pi*Integrate[Exp[-20 t^2]*Sin[t],{t,0,pi}]`.  
Narrow-beam shortcut Gaussian beam,  $\theta_{1/2} \ll 1$

|                                                                                 |  |
|---------------------------------------------------------------------------------|--|
| $\Omega_p \approx \frac{\pi \theta_{1/2}^2}{\ln 2}$ (approximate)               |  |
| Only valid for narrow beams; HW4 P2 is borderline so use the integral.          |  |
| Sanity check $D$ compare to known antennas                                      |  |
| $D < 2$ : short / half-wave dipole. $D \sim 10$ : small horn. $D > 100$ : dish. |  |

RADAR EQUATION (BONUS)

|                                                                                                                               |                                                     |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Monostatic radar                                                                                                              | $tx = rx$ , target RCS $\sigma_t$ (m <sup>2</sup> ) |
| $P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$ (W)                                                                   |                                                     |
| Doppler shift                                                                                                                 | closing velocity $v_r$                              |
| $f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c}$ (Hz)                                                                          |                                                     |
| Factor 2 from round-trip path; $v_r$ radial toward radar. $f_d > 0$ : target approaching. $1/R^4$ falloff vs. Friis $1/R^2$ . |                                                     |

DIPOLE

|                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $l$ — antenna length (m)<br>$\lambda = c/f$ — wavelength (m)<br>$I_0$ — peak current (A)<br>$\eta_0 = 120\pi \approx 377 \Omega$<br>$k = 2\pi/\lambda$ (rad/m)<br>$R_{\text{rad}}$ — rad. res. ( $\Omega$ )<br>$R$ — obs. dist (m) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

RADIATION

|                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $S(R, \theta)$ — pwr dens. (W/m <sup>2</sup> )<br>$S_{\max}$ — peak power density<br>$F(\theta) = S/S_{\max}$ normalized<br>$\Omega_p$ — solid angle (sr)<br>$\beta$ — HPBW (rad or deg)<br>$D$ — directivity (unitless)<br>$P_{\text{rad}}$ — total power (W) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

ANTENNA

|                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_e = \lambda^2 D/(4\pi)$ eff. area<br>$A_p$ — phys. cross sec. (m <sup>2</sup> )<br>$\xi$ — rad. eff. ( $0 < \xi \leq 1$ )<br>$G = \xi D$ — gain (linear)<br>$G_{\text{dB}} = 10 \log G$<br>Half-wave: $D = 1.64$ , $R_{\text{rad}} = 73$<br>Hertzian: $D = 1.5$ |
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FRIIS / LINK

|                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P_t$ — tx power (W)<br>$P_r$ — rx power (W)<br>$G_t, G_r$ — linear gains<br>$R$ — range (m)<br>$L_{fs} = (4\pi R/\lambda)^2$ path loss<br>3 dB $\rightarrow 2$ ; 10 dB $\rightarrow 10$<br>$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$ |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

RADAR

|                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sigma_t$ — RCS (m <sup>2</sup> )<br>$f_d$ — Doppler shift (Hz)<br>$v_r$ — radial vel. (m/s)<br>$P_r \propto 1/R^4$ (monostatic)<br>$f_d = 2v_r/\lambda$ (round-trip)<br>$c = 3 \times 10^8$ m/s<br>$f_d > 0$ : target approaching |
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